1. Types of Applications

Software applications are commonly classified based on **how they are distributed**, **who controls them**, and **what rights the user has**.

Freeware

Freeware is software that is **available at no monetary cost**, but that does **not** mean freedom.

- You can use it for free.
- The source code is **not available**.
- Modification and redistribution are forbidden.
- The developer keeps full control.

BUT: Freeware is often used as bait — free today, restricted tomorrow, or bundled with ads/telemetry.

Open Source Software (By the way)

Open Source software provides access to the **source code**, allowing users to inspect, modify, and distribute it.

- Source code is visible and auditable
- Community-driven development
- Can be free or paid
- License defines redistribution rights

BUT: Open Source ≠ Free Software (philosophically).

Free Software (Libre Software)

Free Software focuses on **user freedoms**, not price:

1. Freedom to run the program
2. Freedom to study how it works
3. Freedom to redistribute
4. Freedom to modify and share modifications

Basically it’s the most free version of software that exists. You can see what it does, control it, modify, see the modifications on your system if it is being installed via package managers. A colaborative type of software since it is usually being maintained by the user community and lives on one-time purchases or software-as-service(if requires connection or updates to work). Often

those projects are being forked, like MySQL and MariaDB, Free Office and Libre Office, Firefox and LibreWolf and Zen Browser, same Debian/Ubuntu and Linux Mint.

Proprietary Software

Proprietary software is **fully controlled by the vendor**.

- Closed-source
- Usage governed by strict licenses
- No modification allowed
- Often tied to hardware or accounts
- Can do a crap like penalties for canceling subscription(see Adobe Photoshop)
- Often redefines agreements, alowly taking off all the rights of the user.

2. Types of Software Licenses

Licenses define **how software can be installed, activated, and transferred**.

OEM License (Original Equipment Manufacturer)

- Comes preinstalled on hardware
- Bound to the original machine
- Cannot legally be transferred

So basically if you modify the software, or just buy a new machine – you loose it forever.

Retail License

- Purchased separately
- Can be transferred between machines
- Usually costs more

Volume License

- Intended for organizations
- One key activates many devices

- Managed via centralized systems
- More expencive at purchase, but cheaper per-unit.

Usually used by: Schools, corporations, governments

3. Boot Managers

A **boot manager** is the software responsible for starting an operating system after firmware initialization (BIOS or UEFI).

NTLDR

- Used by Windows NT / XP
- Legacy BIOS only
- Cannot boot modern systems

Obsolete. Period.

BOOTMGR

- Introduced with Windows Vista
- Supports UEFI
- Uses BCD (Boot Configuration Data)

Still relevant, but Windows-centric only

LILO

- Old Linux bootloader
- No dynamic configuration
- Manual updates required

Dead technology. Historical interest only.

GRUB

- Modular, configurable
- Can boot multiple OSes
- Script-based configuration

GRUB 2

This is **not just an update**, it’s a complete rewrite.

GRUB vs GRUB 2 (Key Differences)

Feature	GRUB	GRUB 2
Codebase	1995	2009
Filesystem support	Limited	Extensive(continuing updates)
UEFI support	No	Yes
Configuration	Static	Dynamic scripts
Extensibility	One masive of code	Modular system
Error recovery	Minimal. Provides tools, but no automatisation or simplifications	Advanced rescue mode(still pretty hard to manage if you're a new user)

4. What is Super Grub2?

Super GRUB2 Disk

Super Grub2 is a **boot recovery tool**, not a bootloader replacement.

Purpose

- Detects installed operating systems
- Boots them **without reinstalling GRUB**
- Repairs broken boot sectors

It exists, because GRUB breaks a lot, especially when you're dualbooting with Windows.

5. EasyBCD?

EasyBCD

EasyBCD is a **Windows-based bootloader configuration tool**.

Functions

- Edits Windows BCD
- Adds Linux/macOS boot entries
- Manages boot order graphically

Limitations

- Depends on Windows BOOTMGR
- Not suitable for pure Linux systems

6. Grub2Win?

Grub2Win

Grub2Win allows **GRUB 2 to run inside Windows**.

Capabilities

- Dual-boot Windows/Linux
- No Linux installation required
- Full GRUB 2 features

Also is being used in WSLs.

7. BootIt?

BootIt

BootIt is a **commercial boot manager and partitioning tool**.

Features

- Multi-boot management
- Advanced partition control
- OS isolation

Used by:

Power users, system administrators, legacy system maintainers.

8. systemd-boot (formerly gummiboot)

systemd-boot

systemd-boot is a **minimal UEFI-only boot manager** developed as part of the **systemd ecosystem**.
It is not a traditional bootloader like GRUB — it does *less*, on purpose.

What it is

- A **UEFI boot manager**, not a full bootloader
- Loads EFI executables directly
- Lives entirely in the **EFI System Partition (ESP)**

No BIOS support. No legacy mode. No excuses.

What systemd-boot is used for

- Booting Linux kernels directly via EFI

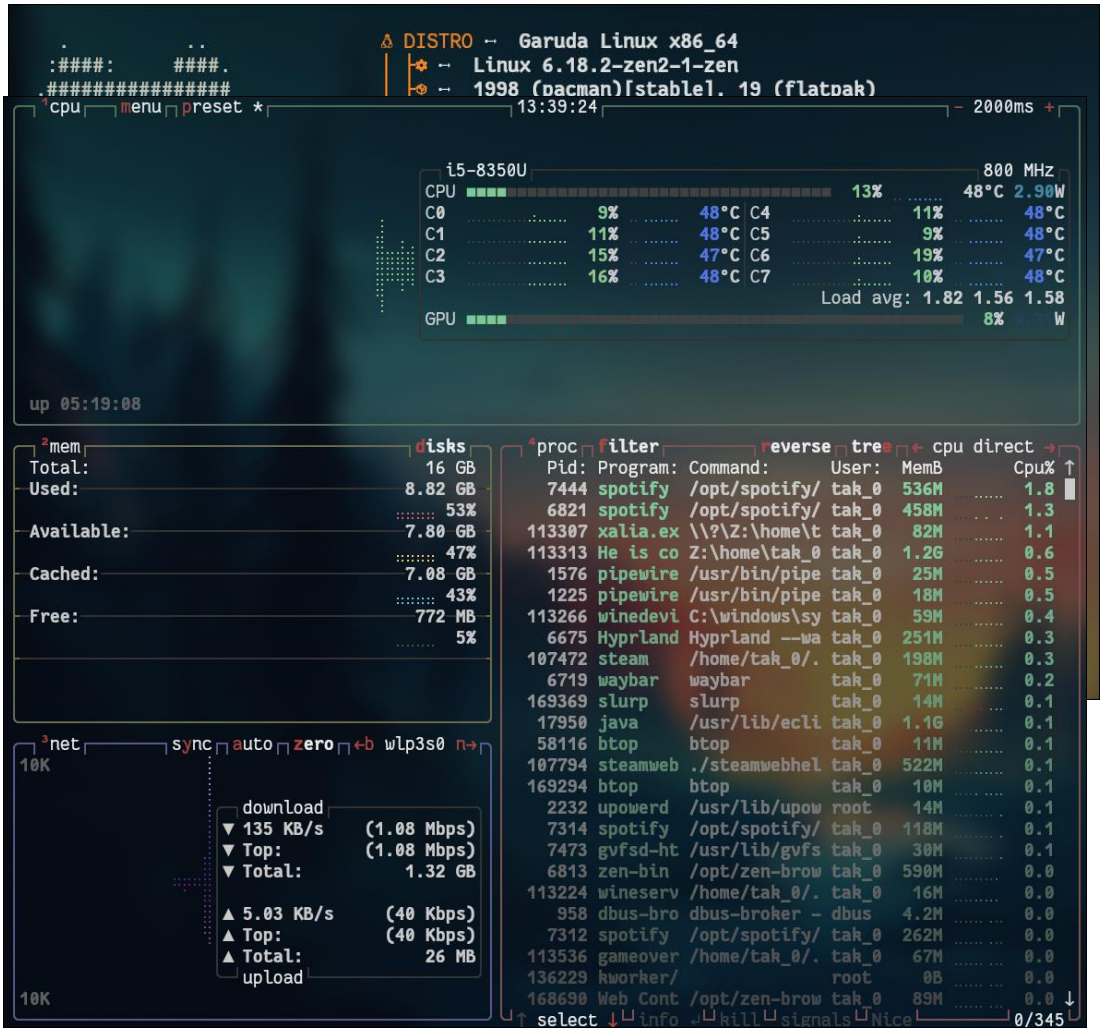
- Managing multiple boot entries cleanly
- Fast, deterministic boot flows
- Systems where **simplicity > flexibility**

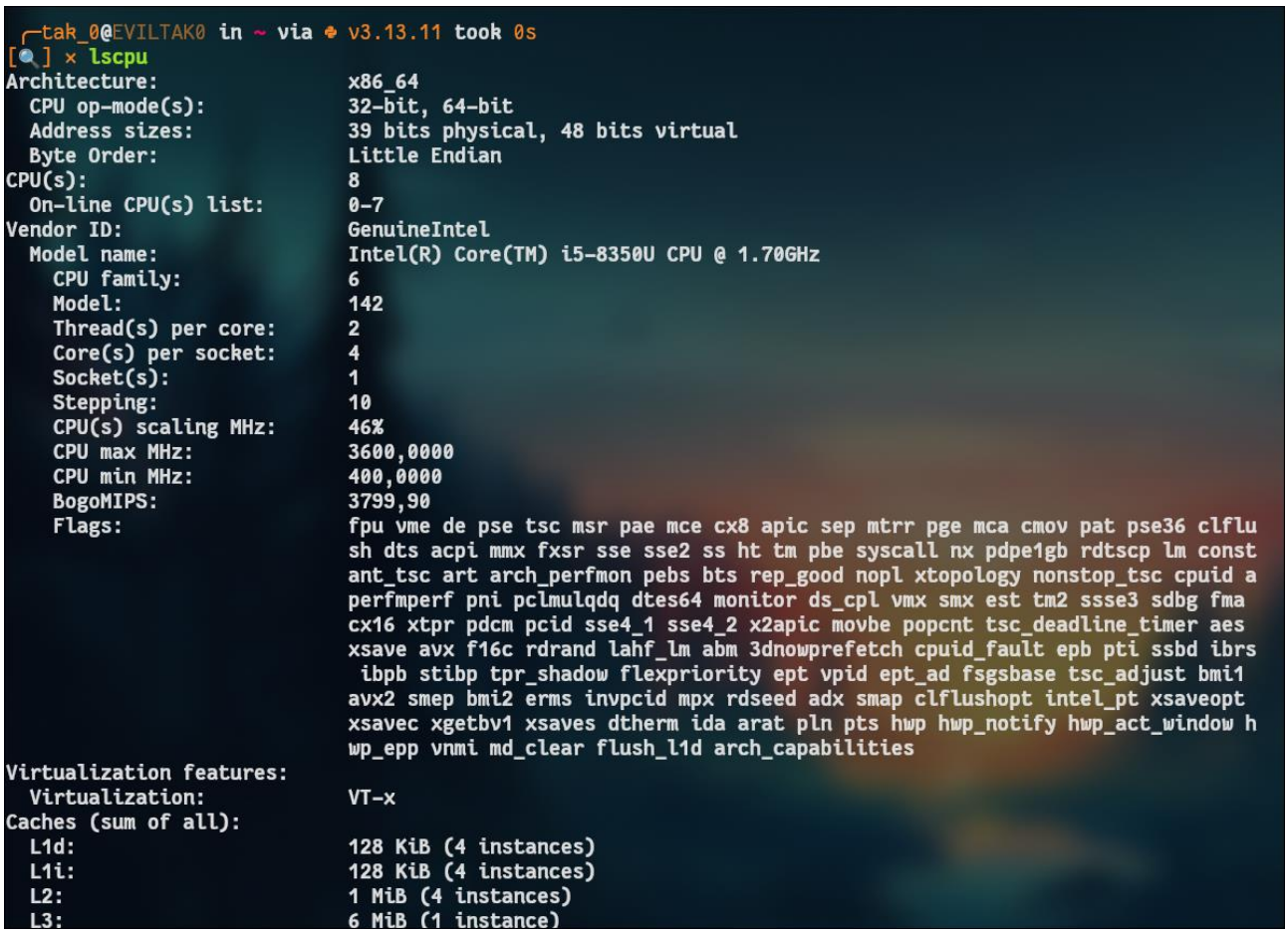
However it has few limitations, such as chainload (loading bootloader from the bootloader), which can be quite important if user manages snapshots, both imperative and declarative OSs at the same time (Ex: Arch Linux + NixOS/GUIX at on the same machine). Also it does not have the filesystem probing.
On the good part – it's way faster than the GRUB.

*** ANEXO *** (Hacer en casa este apartado)
a) No funciona, porque no uso Windows.
Uso Garuda Linux x86_64(portatil + ordenador de casa) con kernel Linux 6.18.2-zen2-1-zen.
16GB(15.49 GiB)

┌─tak_0@EVILTAK0 in ~ via v3.13.11
└─λ uname -a

Linux EVILTAK0 6.18.2-zen2-1-zen #1 ZEN SMP PREEMPT_DYNAMIC Thu, 18 Dec 2025 18:00:58 +0000 x86_64 GNU/Linux




┌─tak_0@EVILTAK0 in ~ via ⬢ v3.13.11 took 0s
[🔍] × lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 8
On-line CPU(s) list: 0-7
Vendor ID: GenuineIntel
Model name: Intel(R) Core(TM) i5-8350U CPU @ 1.70GHz
CPU family: 6
Model: 142
Thread(s) per core: 2
Core(s) per socket: 4
Socket(s): 1
Stepping: 10
CPU(s) scaling MHz: 46%
CPU max MHz: 3600,0000
CPU min MHz: 400,0000
BogoMIPS: 3799,90
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush dts acpi mmx fxsr sse sse2 ss ht tm pbe syscall nx pdpe1gb rdtscp lm constant_tsc art arch_perfmon pebs bts rep_good nopl xtopology nonstop_tsc cpuid aperfmperf pni pclmulqdq dtes64 monitor ds_cpl vmx smx est tm2 ssse3 sdbg fma cx16 xtpr pdcm pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline_timer aes xsave avx f16c rdrand lahf_lm abm 3dnowprefetch cpuid_fault epb pti ssbd ibrs ibpb stibp tpr_shadow flexpriority ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid mpx rdseed adx smap clflushopt intel_pt xsaveopt xsavec xgetbv1 xsaves dtherm ida arat pln pts hwp hwp_notify hwp_act_window hwp_epp vnmi md_clear flush_l1d arch_capabilities
Virtualization features:
Virtualization: VT-x
Caches (sum of all):
L1d: 128 KiB (4 instances)
L1i: 128 KiB (4 instances)
L2: 1 MiB (4 instances)
L3: 6 MiB (1 instance)

```
Caches (sum of all):
  L1d:                    128 KiB (4 instances)
  L1i:                    128 KiB (4 instances)
  L2:                     1 MiB (4 instances)
  L3:                     6 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-7
Vulnerabilities:
  Gather data sampling:   Mitigation; Microcode
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          KVM: Mitigation: Split huge pages
  L1tf:                   Mitigation; PTE Inversion; VMX conditional cache flushes, SMT vulnerable
  Mds:                    Mitigation; Clear CPU buffers; SMT vulnerable
  Meltdown:               Mitigation; PTI
  Mmio stale data:        Mitigation; Clear CPU buffers; SMT vulnerable
  Old microcode:          Not affected
  Reg file data sampling: Not affected
  Retbleed:               Mitigation; IBRS
  Spec rstack overflow:   Not affected
  Spec store bypass:      Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:             Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2:             Mitigation; IBRS; IBPB conditional; STIBP conditional; RSB filling; PBRSB-eIB
                          RS Not affected; BHI Not affected
  Srbds:                  Mitigation; Microcode
  Tsa:                    Not affected
  Tsx async abort:        Mitigation; TSX disabled
  Vmscape:                Mitigation; IBPB before exit to userspace

 ┌─tak_0@EVILTAK0 in ~ via  v3.13.11 took 0s
 └─λ free
               total        used        free      shared  buff/cache   available
Mem:        16239496     8660916      623432     1744072     9038520     7578580
Swap:       34101704      735352    33366352
```

```
 ┌─tak_0@EVILTAK0 in ~ via  v3.13.11 took 0s
 └─λ df /
Filesystem     1K-blocks      Used Available Use% Mounted on
/dev/nvme0n1p2 481935204 188934764 292276308  40% /
```